

Notes for ECE 50877 - Introduction to Applied Cryptography

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Cryptography is an essential component of securing communication and data in a wide range of applications. This course explores fundamental building blocks in cryptography such as ciphers, hash functions, and digital signatures. We will then demonstrate how these building blocks are used practically to secure communications and systems. We will also learn about the power of advanced cryptography to enable complex functionalities with rigorous security guarantees. We will cover fundamental secure multiparty computation protocols, homomorphic encryption, and blockchains, and review their application in building privacy-preserving machine learning. We will also explore social and ethical issues in developing and deploying cryptographic systems.